iv) Herman E. Daly & John B. Cobb Jr, ***For the Common Good: Redirecting the Economy toward Community, the Environment, and a Sustainable Future,*** Beacon Press, Boston, 1989.

iv) Jerry Mander, ***'In the Absence of the Sacred: The Failure of Technology and the Survival of the Indian Nations',*** Sierra Club Books, San Francisco, 1991.)

v) Roberto Vacca, ***'Modest Technologies for a Complicated World',*** Pergamon Press, New York, 1980.

13. The world economy is consuming resources and generating wastes at unprecedented rates. In the past 100 years, the world's industrial production increased more than 50-fold. See W.W. Rostow, 'The World Economy: History and Prospects', University of Texas Press, Teaxas, 1978, pp. 48-49.

14. Clive Ponting, 'Historical Perspectives on Sustainable Development', Environment, Vol. 32, No. 9, November 1990.

15. James Galloway, et al. Atmospheric Environment 16 (7): 1678, 1982. Also see Robert U. Ayres, ***'Toxic Heavy Metals: Materials Cycle Optimization',*** Proceedings of the National Academy of Sciences, vol. 89, No. 3, Feb. 1, 1992, pp. 815-820.

16. U.S. Congress, Office of Technology Assessment, ***Changing By Degrees: Steps to Reduce Greenhouse Gases,*** OTA-O-482, U.S. Government Printing Office, Washington, DC, February 1991.

17. Claude de Levi-Strauss, 'The Savage Mind', Black Swan, Paris, 1985. The 'savage mind' has been used by him to prove the 'uncivility' of not only the tribal people but a large segment of the developing societies of today's times. The French philosopher is also credited to develop the idea of 'structuralism'.

18. For example, advances in agricultural science have greatly improved food production around the world, but at the same time the use of pesticides, an integral part of modern agricultural methods, have led to a number of environmental and human health problems. On the other hand, new pollution control technologies have greatly reduced environmental emissions, while new manufacturing technologies have improved materials and energy efficiency. From 1972 to 1985, for instance, the Gross Domestic Product (GDP) of the United States grew by about 40 percent, but energy consumption remained basically flat. Nearly two-thirds of this decline in energy consumption was due to the introduction of energy efficient technologies; the remaining third was due to structural shifts in the composition of the economy (i.e., a shift away from heavy manufacturing towards services). See U.S. Congress, Office of Technology Assessment, ***Energy Use and the U.S. Economy,*** OTA-BP-E-57, U.S. Government Printing Office, Washington, DC, June 1990.

19. See U.S. Congress, Office of Technology Assessment, ***'Green Products By Design: Choices for a Cleaner Environment',*** OTA-E-541 (Washington, DC: U.S. Government Printing Office, October 1992).

20. The discussion here is based on the taxonomy developed by Colby. A set of five paradigms, ranging from 'frontier economics' to 'deep ecology' is used describe the relationship between economic development and the environment. One extreme, frontier economics, focuses on economic growth and emphasises free markets and unbridled exploitation of resources. The other extreme, deep ecology, focuses on harmony with nature and emphasises drastic reductions in human population and the scale of human economies. Here we describe the technology implications of the three middle paradigms. See Michael E. Colby,